

EDUCATION

- University of Edinburgh** Edinburgh, UK
• *PhD in Computer Science* *August 2023 - Present*
Research Field: AI-Systems, Distributed ML, ML-oriented Architecture, Serverless
- University of Science and Technology of China** Hefei, China
• *Bachelor of Computer Science; The school of Gifted Young;* *July 2017 - June 2021*
Courses: Operating Systems, Artificial Intelligence, Principles of Compiler, Computer Architecture, High Performance Architecture

PUBLICATIONS

- Yeqi Huang**, Congjie He, Haocheng Xiao, et al. "Wavel: A Fast and Efficient Compilation System for Wafer-Scale Accelerators." *Accepted to the ACM Symposium on Operating Systems Principles (SOSP)*, 2026.
- Congjie He, Le Xu, Zhan Lu, **Yeqi Huang**, et al. "MeshRT: Compile-Time Governed Wafer-Scale Runtime for Low-Latency High-Throughput Inference." *Accepted to the ACM Symposium on Operating Systems Principles (SOSP)*, 2026.
- Yeqi Huang**, Yanwei Ye, Guomin Chen, et al. "SwarmPilot: A Scheduler Agent Framework for Large Agentic Workflow Clusters." *Accepted to the European Conference on Computer Systems (EuroSys)*, 2026.
- Yeqi Huang**, Yue Chen, Yanwei Ye, et al. "BioVLM: Evidence-Enriched Data Synthesis from Biomedical Papers." *Under Review, ACL 2025 System Demonstration.*
- Chuanming Zha, Jiamin Zheng, **Yeqi Huang**, et al. "LLM-Monitor: Efficient Privacy Violation Monitoring for LLMs."
- Yinsicheng Jiang*, **Yeqi Huang***, Liang Cheng, et al. "ContextPilot: Efficient Retrieval-Augmented Generation with Accuracy-Preserving Context Reuse." *MLSys*, 2026. (*Equal Contribution)
- Congjie He, **Yeqi Huang**, Pei Mu, et al. "WaferLLM: Large Language Model Inference at Wafer Scale." *Under Review, OSDI 2025.*
- Yao Fu*, Yinsicheng Jiang*, **Yeqi Huang***, et al. "MoE-CAP: Cost-Accuracy-Performance Benchmarking for Mixture-of-Experts Systems." *NeurIPS*, 2025. (*Equal Contribution)
- Yao Fu, Leyang Xue, **Yeqi Huang**, et al. "ServerlessLLM: Locality-Enhanced Serverless Inference for Large Language Models." In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 2024.
- Xiao-Long Chen, Lin-Feng Wang, **Yeqi Huang**, et al. "Symplectic structure-preserving particle-in-cell whole-volume simulation of tokamak plasmas." In *SC21: International Conference for High Performance Computing, Networking, Storage and Analysis*, 2021.

PROJECTS

- Wavel (SOSP 2026)**: Designed MeshIR and locality-preserving schedule search for wafer-scale accelerators, achieving up to $3.6\times$ speedup and reducing search from weeks to seconds.
- MeshRT (SOSP 2026)**: Compiled dynamic LLM inference events into decentralized per-core schedules, improving throughput by $30\text{--}68\times$ while preserving ultra-low latency.
- SwarmPilot (EuroSys 2026)**: Developed scheduler agents with neural predictors, memory, tools, and coordination; production deployment improved P99 latency by up to 50% and doubled throughput.
- BioVLM**: Built an evidence-enriched biomedical VLM data pipeline and applied SFT and RL to Qwen3-VL-8B, outperforming GPT-5.2 on LAB-Bench at under \$200 training cost.
- LLM-Monitor**: Compiled privacy policies into efficient local-model monitors, matching GPT-5.2 monitoring quality with a specialized Qwen3-8B model at $15\times$ higher speed.
- AutoReader**: Built graph and vector retrieval for scientific literature, with $60\times$ faster indexing than NVIDIA cuVS and $13\times$ faster retrieval than Milvus.

- **AI4Math - Sketchpad:** Building an AI for Math-funded system that converts proofs into structured graphs and decomposes them for precise auto-formalization.
- **QED Simulation on GPU:** Implemented CUDA Monte Carlo simulation with 1500–5000× speedup; recognized by Nobel Laureate Frank Wilczek at APS.

TALKS

- 1 **Yeqi Huang.** "InfiniTensor: A Tensor-Friendly, Efficient Parallel Programming Library for Accelerator-Centric Clusters." *UKSys 2024*.
- 2 **Yeqi Huang.** "Why we need a new clustering benchmark in AI retrieval?" *International Workshop on Efficient Generative AI*, 2024.

TEACHING EXPERIENCE

- **Machine Learning System TA (2025):**
 - Teaching assistant for CUDA programming course, covering GPU architecture, CUDA, CuPy, and Triton.
 - Designed hands-on exercises and projects to help students master parallel programming concepts.
- **Machine Learning System TA (2026):**
 - Teaching assistant with new CuTile programming module, teaching Tile-based GPU programming.

AWARDS

International Awards:

- International Supercomputing Conference Student Cluster Competition Champion - 2021
- Asian Supercomputer Conference Student Cluster Competition First Prize - 2021

National Awards:

- Best Chinese Supercomputing Application of the Year - 2022
- Huawei Pioneer Developer (4 in China) - 2021
- National Compiler Designing Competition Second Prize - 2021

SKILLS

- **LLM Systems & Inference:** High-performance serving, distributed and wafer-scale inference, batching and MoE runtime support, model loading and migration, KV/context reuse, and cross-hardware optimization on GPUs, Apple Silicon, and Cerebras.
- **LLM Training & Evaluation:** Evidence-grounded data synthesis; SFT and RL post-training; inference-time scaling; quality, latency, throughput, energy, and cost evaluation for LLM and VLM systems.
- **RAG & Agents:** Vector and graph retrieval, context indexing/deduplication, multi-agent workflows, neural prediction, memory and tool abstractions, and production integration with Ray and ComfyUI.
- **Systems & Programming:** C++, Python, Rust; CUDA, CuPy, Triton, CuTile, MLIR/LLVM, OpenMP, Linux/eBPF, distributed systems, compiler tooling, and performance engineering.
- **Physics & Math:** Numerical methods, linear algebra, computational fluid and molecular dynamics, quantum mechanics, and quantum electrodynamics.

SPECIAL EXPERIENCE

- **Open Source Enthusiast:** Contributing to notable projects like GNOME, LAMMPS, and LLVM, I've used GitHub since 2019 to showcase my development journey and ideas.
- **Running a Science-Themed Cafe:** Leveraging software development income, I established Quantum Coffee near my school—a collaborative space encouraging students to explore and discuss scientific topics across disciplines.
- **UNICEF Charity Projects:** I have donated 10% of all personal income to children's charities since 2019.